

11 Eigenvalue Decomposition



11.2 Spectral Decomposition

A real symmetric matrix's eigendecomposition is geometrically "rotate $\rightarrow$ scale $\rightarrow$ rotate" — spectral decomposition finds a coordinate system that straightens an ellipse

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master in This Section

1

A real symmetric matrix satisfies $A=A^T$, with all real-valued entries

2

A real symmetric matrix's spectral decomposition reads as "rotate $\rightarrow$ scale $\rightarrow$ rotate"

3

Spectral decomposition finds a new coordinate system that straightens an ellipse

4

The unit circle becomes a rotated ellipse — this reveals the geometric meaning of eigenvalues

5

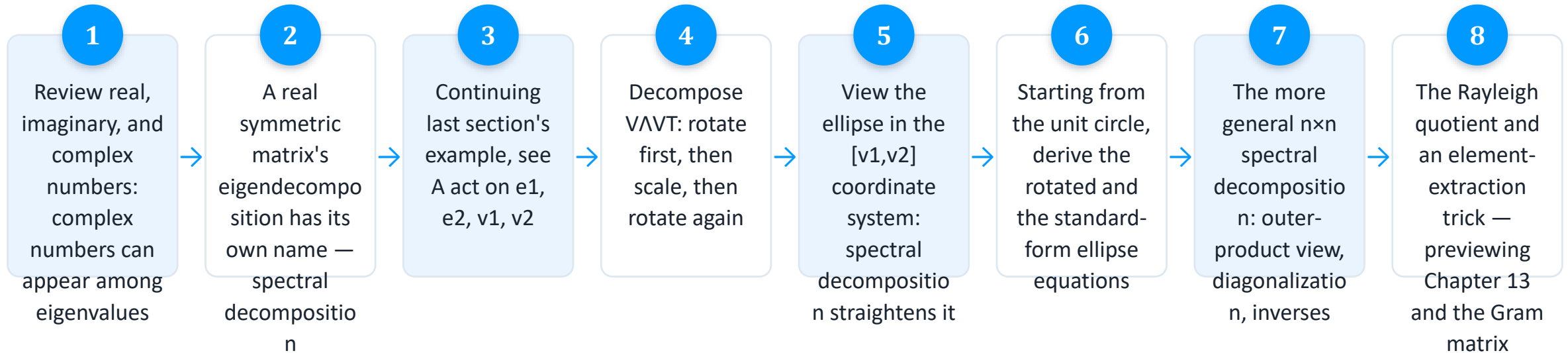
A real symmetric matrix is invertible exactly when all its eigenvalues are nonzero

6

The Rayleigh quotient measures a vector's scaling under a matrix — a preview of Chapter 13

SECTION STRUCTURE

From $A=V\Lambda V^T$ to General Spectral Decomposition, Ellipses, and Inverses



Reviewing Real, Imaginary, and Complex Numbers

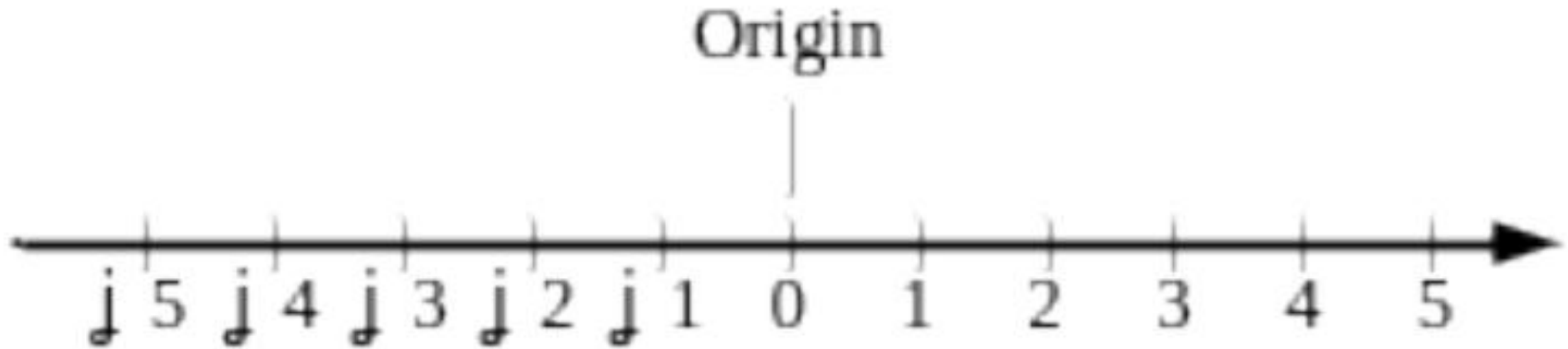
Real numbers: rational numbers (like $1/3$) and irrational numbers (like π); every real number sits at a point on the number line

Imaginary numbers: extend square roots to negative numbers; the imaginary unit i satisfies $i^2 = -1$

Complex numbers: written $a+bi$ (a, b real); eigenvalues and eigenvectors can be complex, though complex-matrix operations are not this book's topic

Figure 1 · The Real Number Line

- Every real number corresponds to a point on the number line, and vice versa



For an $n \times n$ real symmetric matrix A , its eigendecomposition can be written

$$A = V\Lambda V^T$$

- V is orthogonal, satisfying $VV^T = V^T V = I$; V 's columns are A 's eigenvectors
- Λ is diagonal, with A 's eigenvalues on the main diagonal — this decomposition is called spectral decomposition

The 2×2 real symmetric matrix A's spectral decomposition, with numbers substituted

$$A = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

- $V=[v_1, v_2]$ is a rotation matrix; $\Lambda=\text{diag}(2, 1/2)$ is a scaling matrix

Figure 2 · Matrix A's Linear Transform of e_1, e_2

- The unit square grid shears into a parallelogram; the unit circle becomes a rotated ellipse

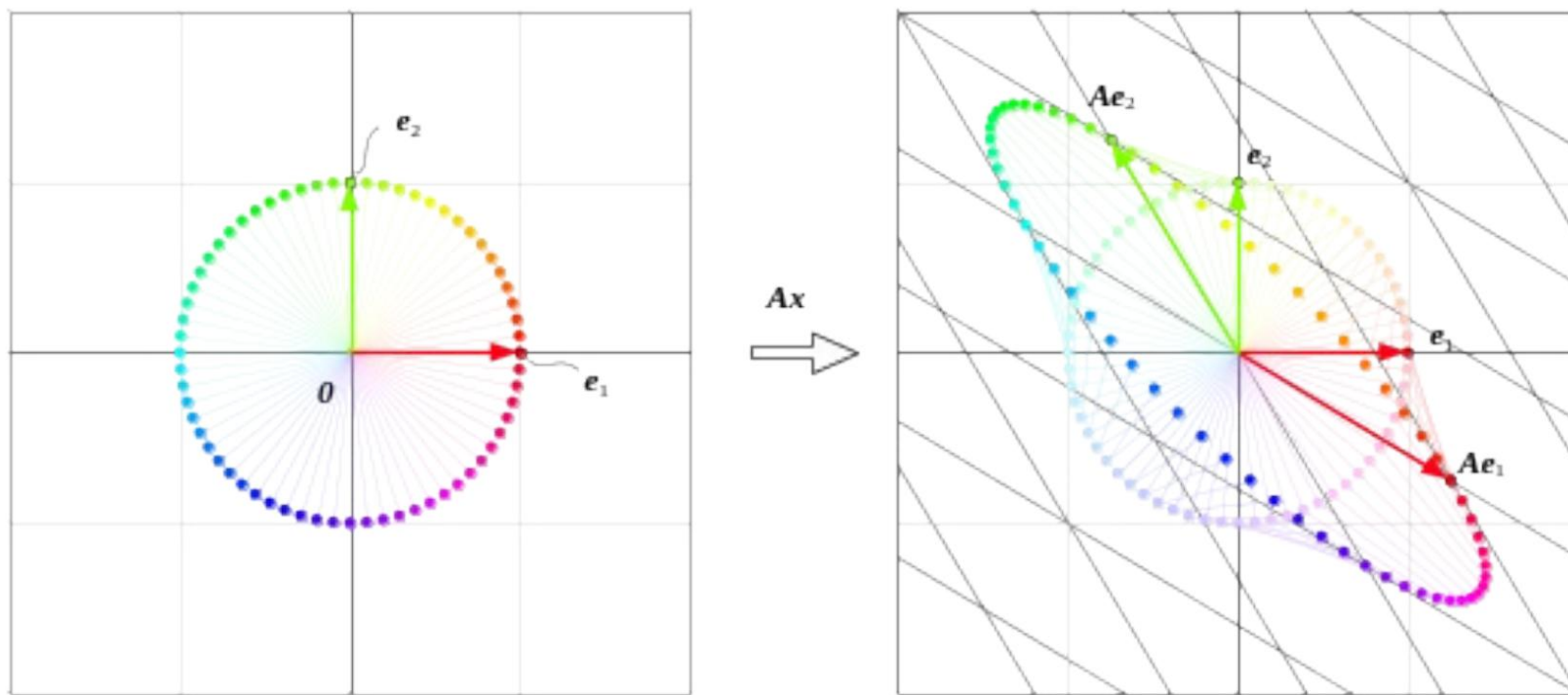


图 2. 矩阵 A 对 e_1, e_2 的线性变换

Figure 3 · Matrix A's Linear Transform of v_1, v_2

- v_1, v_2 are A's eigenvectors — their direction stays fixed under the transform, only stretched

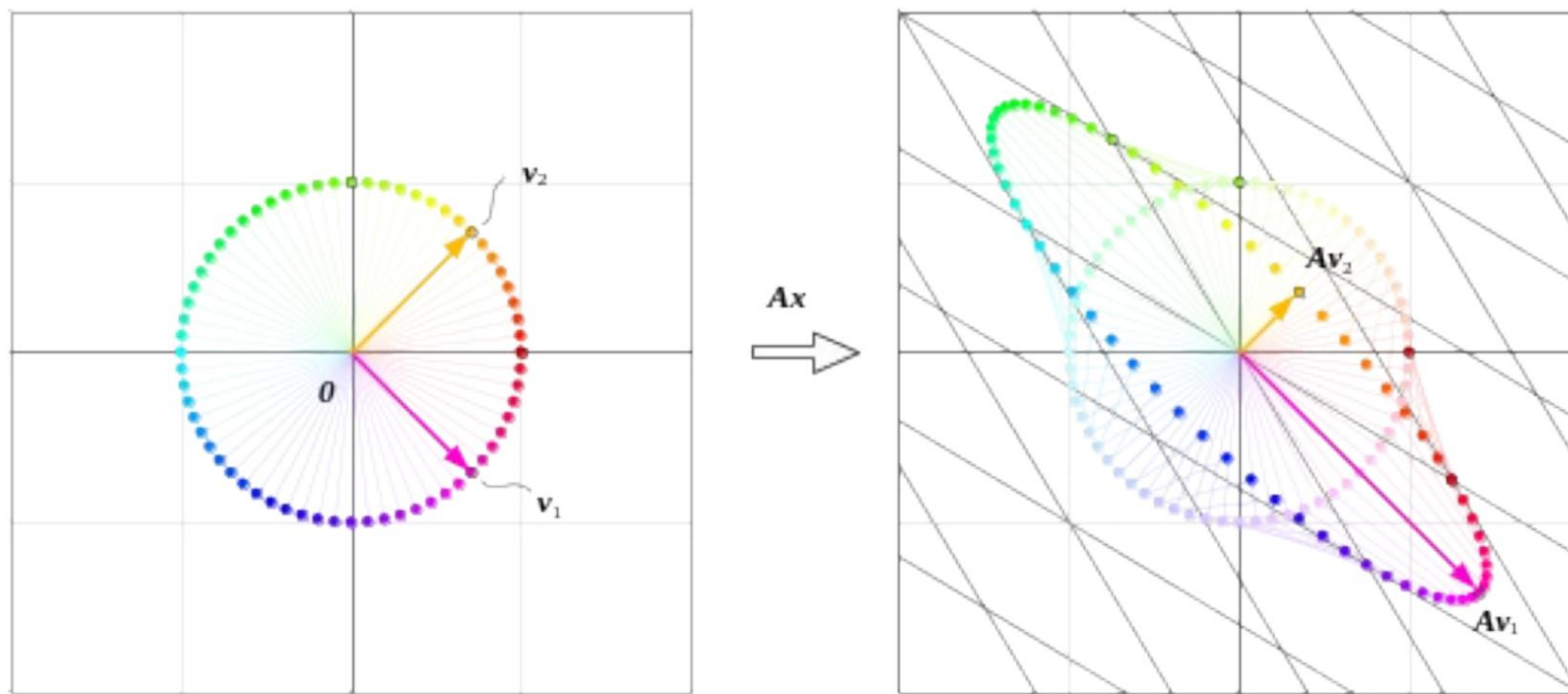


图 3. 矩阵 A 对 v_1, v_2 的线性变换

A can be viewed as three consecutive actions: rotate, then scale, then rotate again

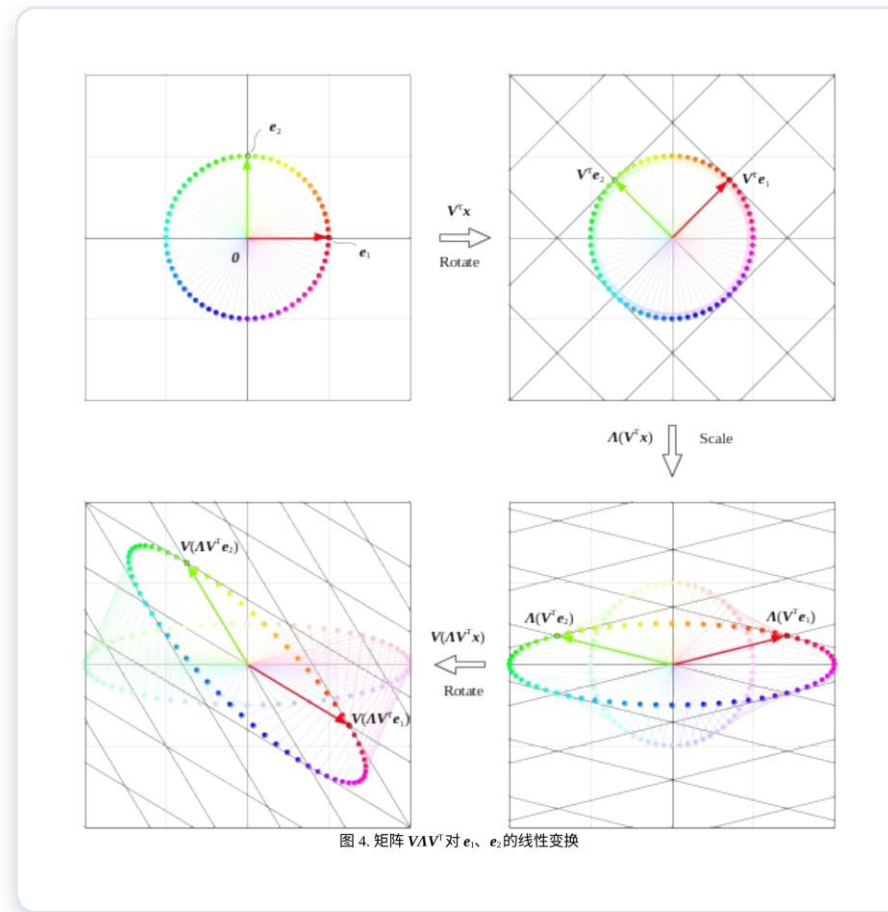
Step 1: rotate with V^T — for example $V^T e_1, V^T e_2$; V^T 's determinant is 1, so rotation preserves area

Step 2: scale with $\Lambda = \text{diag}(2, 1/2)$ — 2x on the horizontal axis, 1/2x on the vertical; Λ 's determinant is also, coincidentally, 1

Step 3: rotate again with V to reach the final result; A 's determinant equals the product of all its eigenvalues

Figure 4 · Matrix $V\Lambda V^T$'s Linear Transform of e_1, e_2

- Tracking e_1, e_2 step by step makes the "rotate → scale → rotate" sequence visible



V1, V2: ROTATE → SCALE → ROTATE

For eigenvectors v_1, v_2 , the VT -rotation result is quite special

$VTv_1 = [1, 0]^T = e_1$, $VTv_2 = [0, 1]^T = e_2$ — not a coincidence, but a result of V 's columns being orthogonal

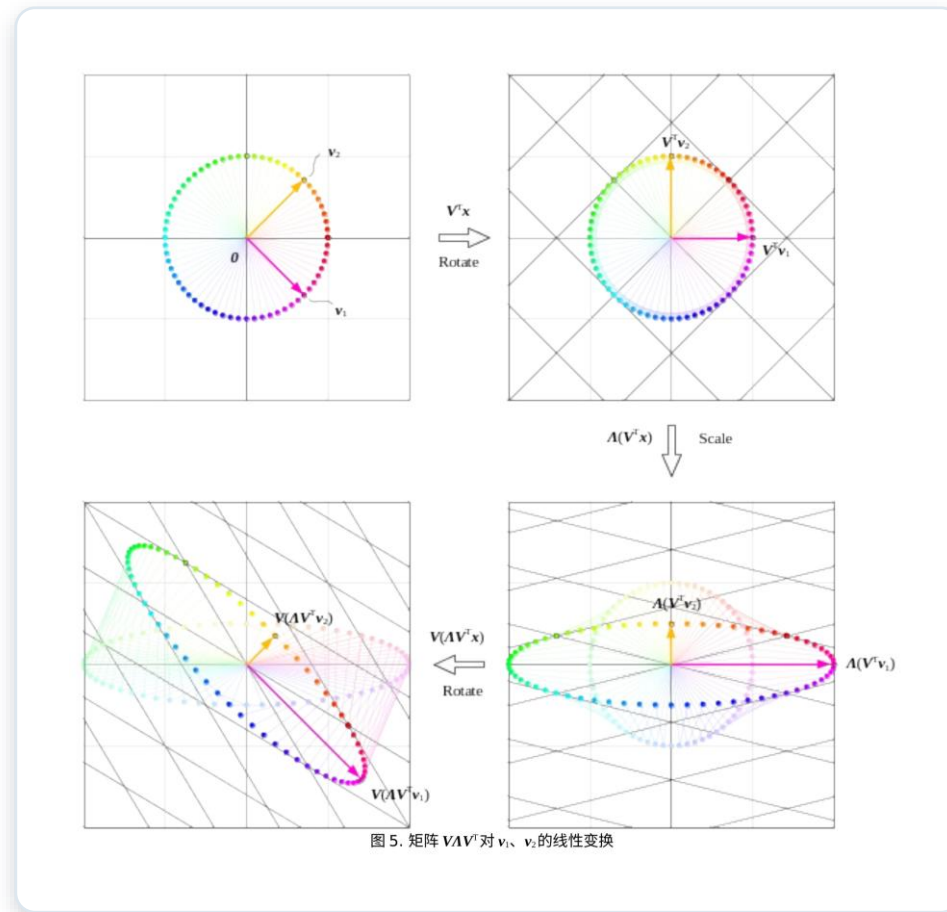
Scaling VTv_1, VTv_2 by Λ turns the unit circle into a standard-form ellipse

Rotating with V last: $V(\Lambda VTv_1) = 2v_1$, $V(\Lambda VTv_2) = 0.5v_2$ — the standard ellipse rotates into the rotated ellipse

V1, V2: ROTATE → SCALE → ROTATE

Figure 5 · Matrix $V\Lambda V^T$'s Linear Transform of v_1, v_2

- $V(\Lambda V^T v_1) = 2v_1$, $V(\Lambda V^T v_2) = 0.5v_2$ — confirming v_1, v_2 are indeed A 's eigenvectors



VIEWING THE ELLIPSE IN $[v_1, v_2]$

Geometrically, spectral decomposition finds a coordinate system that straightens the ellipse

V 's columns form an orthonormal basis; viewed in $[v_1, v_2]$, the rotated ellipse becomes a standard-form ellipse λ_1, λ_2 are exactly the ellipse's semi-axis lengths

The next chapter's topic, principal component analysis (PCA), builds on exactly this idea

Figure 6 · Viewing the Ellipse in $[e_1, e_2]$ and $[v_1, v_2]$

- The left ellipse is rotated; switching to the $[v_1, v_2]$ system, the right ellipse is "straightened"

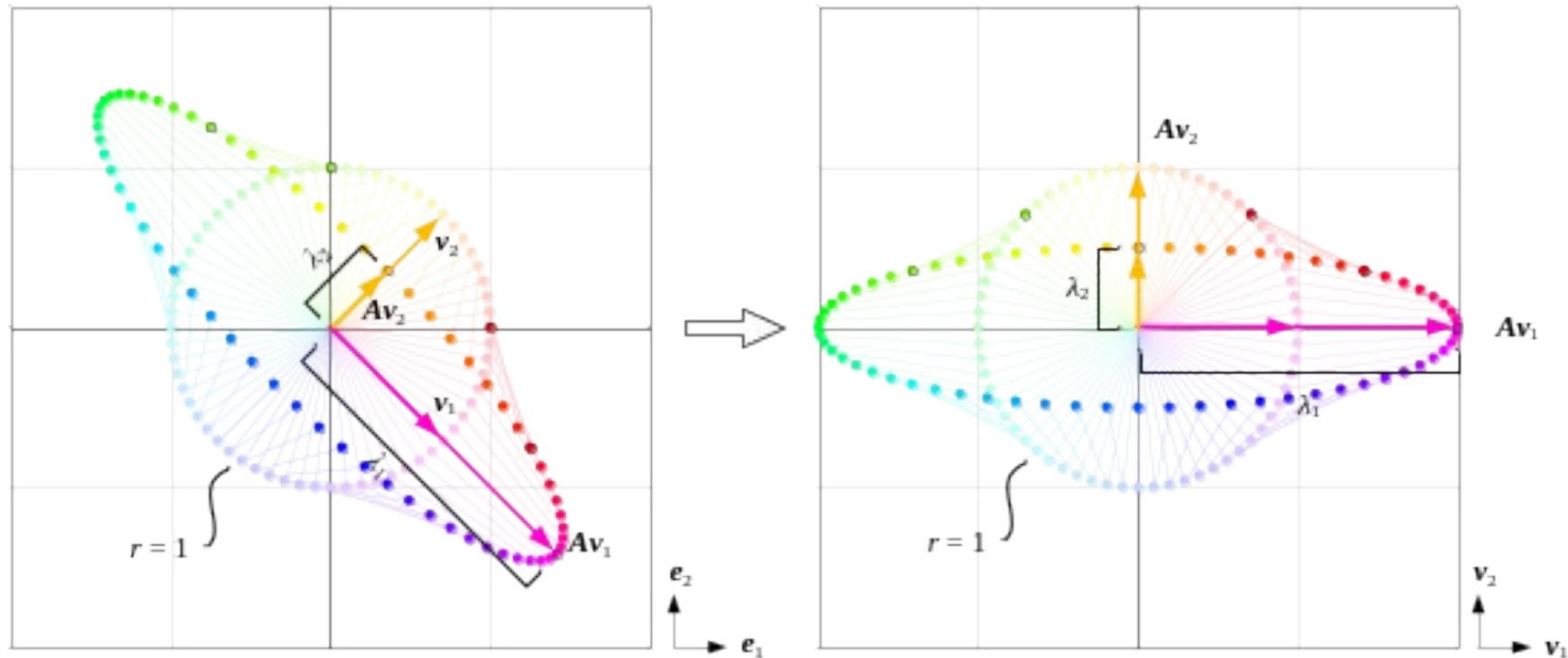


图 6. 分别在 $[e_1, e_2]$ 、 $[v_1, v_2]$ 中观察椭圆

ELLIPSES

An ellipse is a conic section: the locus of points whose distances to two foci sum to a constant

The two fixed points are foci, F_1 and F_2 ; any point P on the ellipse satisfies $PF_1 + PF_2 = 2a$

The major axis passes through both foci with length $2a$ (a is the semi-major axis); the minor axis is perpendicular, length $2b$ (b is the semi-minor axis)

An ellipse with axes parallel to the coordinate axes is a standard-form ellipse; rotating it about its center gives a rotated ellipse

Figure 7 · An Ellipse Centered at the Origin, Foci on the Horizontal Axis

- $a > b > 0$; the ellipse equation $x^2/a^2 + y^2/b^2 = 1$ is just as compact in matrix multiplication form

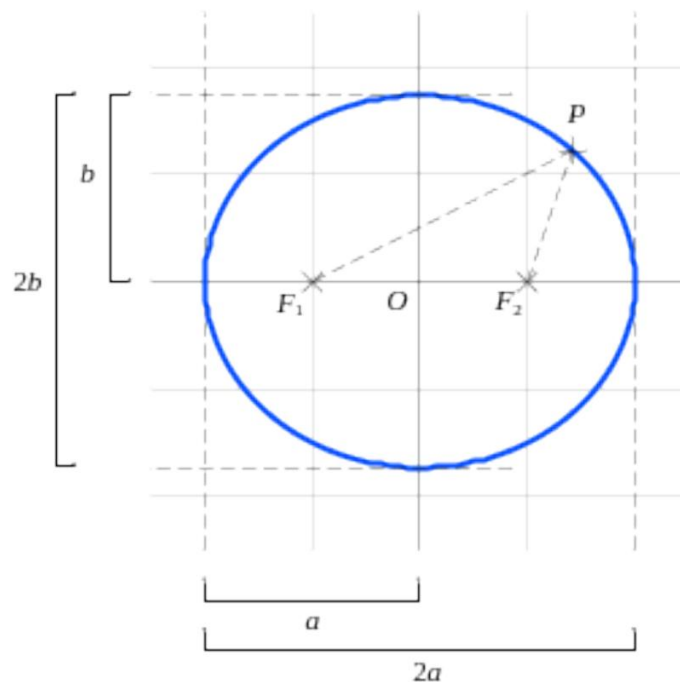


图 7. 中心位于原点，焦点位于横轴的椭圆

A unit vector $\mathbf{z}$ is transformed by $\mathbf{A}$ into $\mathbf{x}$, $\mathbf{A}\mathbf{z}=\mathbf{x}$; solving $\mathbf{z}=\mathbf{A}^{-1}\mathbf{x}$ and substituting into $\|\mathbf{z}\|^2=1$

$$\mathbf{x}^T(\mathbf{A}^{-1})^T\mathbf{A}^{-1}\mathbf{x} = 1$$

- Substituting this section's $\mathbf{A}^{-1}$ and simplifying gives the rotated ellipse's equation, $2.125x_1^2+3.75x_1x_2+2.125x_2^2=1$
- Reading off the semi-major and semi-minor axes directly from this form is not easy

Since $A^{-1}=V\Lambda^{-1}V^T$, substituting $y=V^Tx$ and simplifying gives the standard-form ellipse

$$y_1^2/\lambda_1^2 + y_2^2/\lambda_2^2 = 1$$

- This is the standard-form ellipse we wanted — the semi-major axis λ_1 and semi-minor axis λ_2 are now plain to see
- Besides ellipses, conic sections also include hyperbolas and parabolas — their link to linear algebra is Chapter 13's topic

For an $n \times n$ real symmetric matrix A , the spectral decomposition takes the same form

$$A = V\Lambda V^T$$

- Λ 's diagonal entries run largest to smallest, top-left to bottom-right; geometrically, V rotates and Λ scales

Figure 8 · $n \times n$ Real Symmetric Matrix A 's Spectral Decomposition

- V 's columns are eigenvectors; Λ 's diagonal holds the eigenvalues, from λ_1 down to λ_n

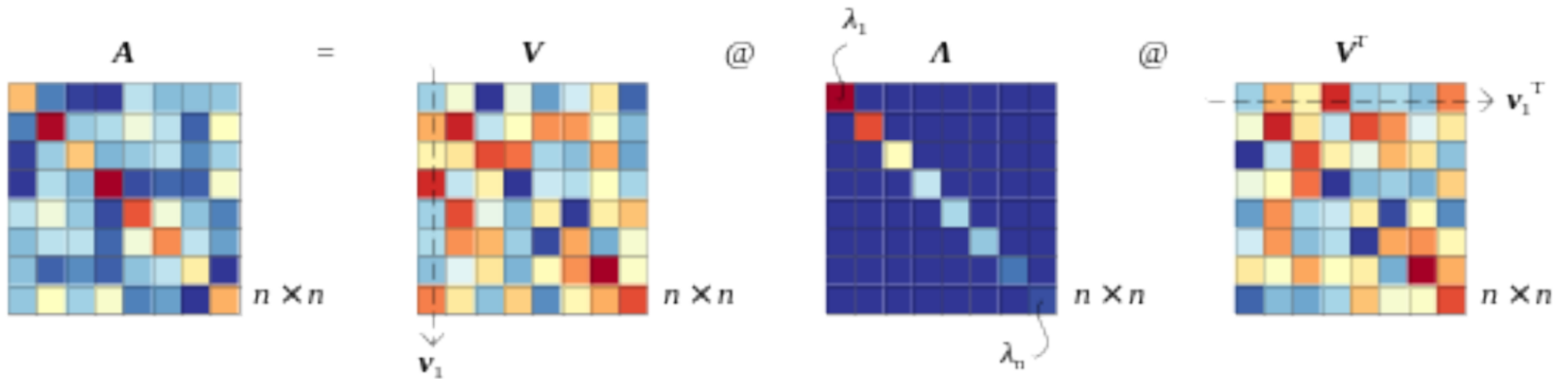


图 8. $n \times n$ 实对称矩阵 A 谱分解

In the outer-product view, $A=V\Lambda V^T$ can be written as a sum of n rank-one matrices

$$A = \lambda_1 \mathbf{v}_1 \mathbf{v}_1^T + \lambda_2 \mathbf{v}_2 \mathbf{v}_2^T + \cdots + \lambda_n \mathbf{v}_n \mathbf{v}_n^T$$

- Larger eigenvalues give more "vivid" sub-images; as long as $\lambda_i \neq 0$, $\lambda_i \mathbf{v}_i \mathbf{v}_i^T$ is a rank-one matrix

Figure 9 · A Sum of n Matrices Gives A

- Spectral decomposition "disassembles" A into these sub-matrices, like light split into color bands



图 9. n 个矩阵的叠加得到 A

Since V is invertible, diagonalizing the real symmetric matrix A corresponds to

$$V^T A V = \Lambda$$

- This is exactly "straightening" A into the diagonal matrix Λ , viewed in the orthonormal basis formed by V 's columns

Figure 10 · $n \times n$ Real Symmetric Matrix A 's Diagonalization

- Expanding VTAV: the main-diagonal entries are eigenvalues, the off-diagonal entries are 0

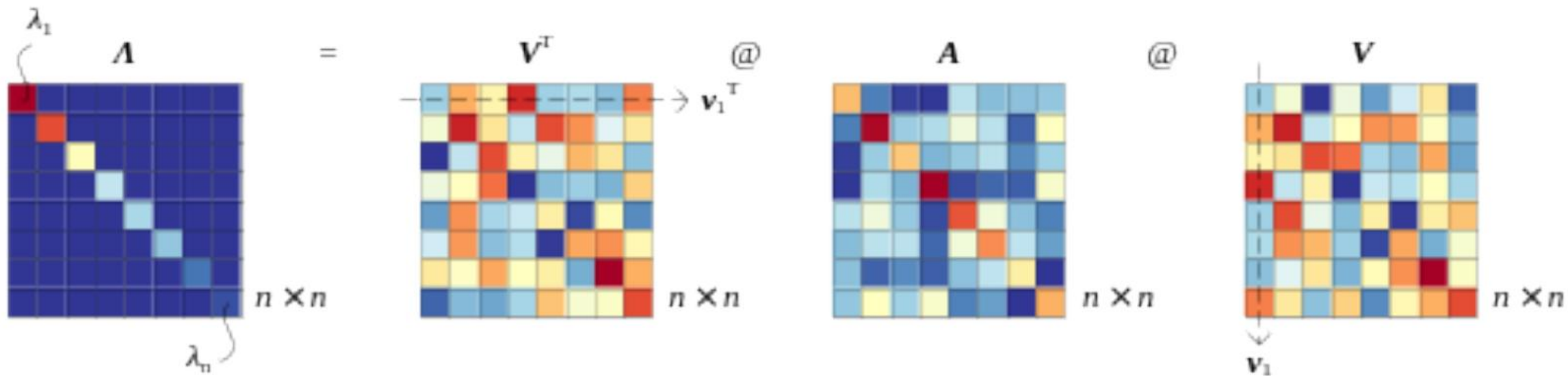


图 10 · $n \times n$ 实对称矩阵 A 的对角化

Take $\mathbf{v}^T A \mathbf{v} = \lambda$ as an example — since $A\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{v}$ is a unit vector

$$\mathbf{v}^T A \mathbf{v} / \mathbf{v}^T \mathbf{v} = R(\mathbf{v})$$

- $R(\mathbf{x})$ is called the Rayleigh quotient, measuring how much $\mathbf{x}$ is scaled under A ; Chapter 13, Section 3 covers it in full

Take $\mathbf{v}_2^T \mathbf{A} \mathbf{v}_1 = 0$ as an example — since V 's columns are pairwise orthogonal

$\mathbf{v}_2^T \mathbf{A} \mathbf{v}_1 = \lambda_1 \mathbf{v}_2^T \mathbf{v}_1 = 0$, showing that eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ stay orthogonal under this particular linear transform

Since A is real symmetric, transposing gives the same result:
 $\mathbf{v}_1^T \mathbf{A} \mathbf{v}_2 = \lambda_1 \mathbf{v}_1^T \mathbf{v}_2 = 0$

Figures 11, 12 · Extracting Λ 's Main-Diagonal and Off-Diagonal Entries

- Left: e1TVTAVe1 extracts Λ 's row 1, column 1 (eigenvalue λ_1); right: e2TVTAVe1 extracts row 2, column 1 (which is 0)

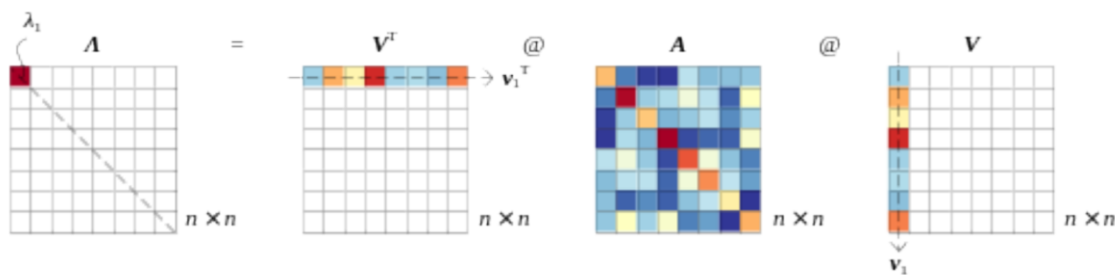


图 11. $n \times n$ 实对称矩阵 A 对角化，主对角线元素

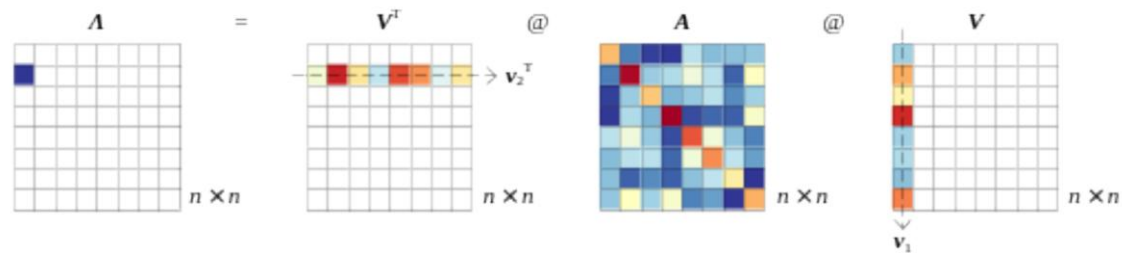


图 12. $n \times n$ 实对称矩阵 A 对角化，非主对角线元素

A real symmetric matrix A has an inverse only when all its eigenvalues are nonzero

$$A^{-1} = V\Lambda^{-1}V^T$$

- This is exactly the spectral decomposition of A^{-1} : the eigenvectors stay the same, the eigenvalues are inverted

Figure 13 · $n \times n$ Real Symmetric Matrix A^{-1} 's Spectral Decomposition

- Λ^{-1} 's diagonal runs from $1/\lambda_1$ to $1/\lambda_n$, provided every eigenvalue is nonzero

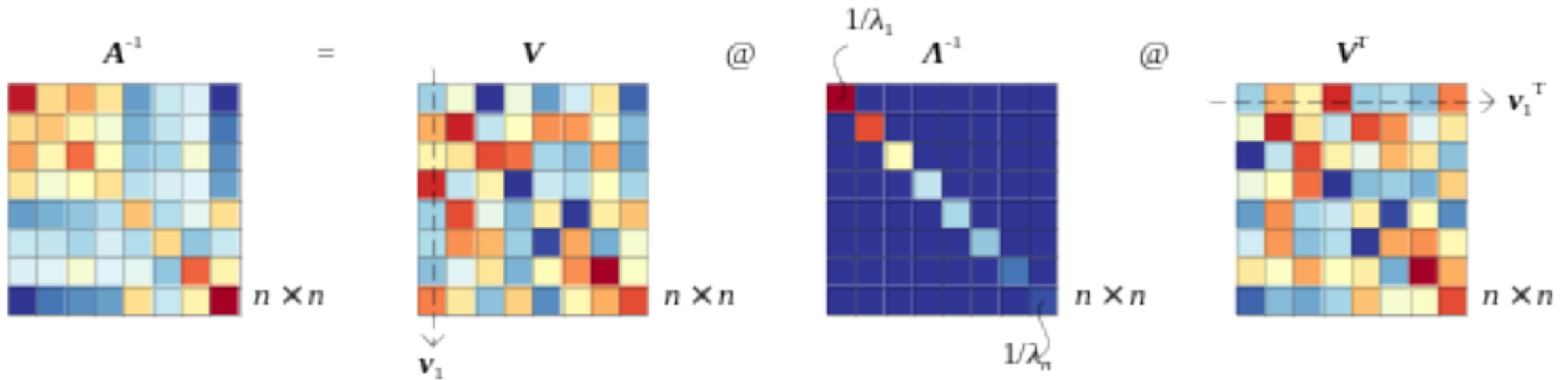


图 13. $n \times n$ 实对称矩阵 A^{-1} 谱分解

$e_i^T l e_j$ extracts an identity-matrix entry; $v_i^T A v_j$ extracts a diagonal-matrix Λ entry

$e_1^T l e_1 = 1$, $e_2^T l e_1 = 0$ — these simply extract the corresponding entry of the identity matrix I

$v_1^T A v_1 = \lambda_1$, $v_2^T A v_1 = 0$ — these extract the corresponding entry of Λ (not A), viewed in the orthonormal basis V forms

The most common symmetric matrix seen earlier in this book is the Gram matrix; this chapter's Section 4 covers it in full

SUMMARY

What Have We Seen About Spectral Decomposition?

Spectral decomposition is a real symmetric matrix's eigendecomposition, $A = V\Lambda V^T$, with V orthogonal and Λ diagonal

Geometrically: spectral decomposition is "rotate $\rightarrow$ scale $\rightarrow$ rotate"

It finds a coordinate system $[v_1, v_2]$ that straightens a rotated ellipse into standard form

λ_1, λ_2 are both the eigenvalues and the straightened ellipse's semi-axis lengths

Outer-product view: $A = \sum \lambda_i v_i v_i^T$, decomposing A into a sum of n rank-one matrices

A real symmetric matrix A is invertible only when all its eigenvalues are nonzero, $A^{-1} = V\Lambda^{-1}V^T$

Key Terms in This Section

Spectral Decomposition *A real symmetric matrix's eigendecomposition, $A=V\Lambda V^T$*

Orthogonal Matrix V *Satisfies $VV^T=V^TV=I$; geometrically represents a rotation*

Standard-Form / Rotated Ellipse *Axes parallel to the coordinate axes is standard-form; rotated about the center is a rotated ellipse*

Diagonalization *$V\Lambda V^T=A$ — "straightening" A in the orthonormal basis V forms*

Rank-One Matrix *A matrix of the form $\lambda_i v_i v_i^T$; rank 1 whenever λ_i is nonzero*

Real Symmetric Matrix *A square matrix satisfying $A=A^T$ with all real-valued entries*

Ellipse *The locus with constant sum of distances to two foci; semi-major axis a , semi-minor axis b*

Outer-Product View *Writing $A=V\Lambda V^T$ as $\lambda_1 v_1 v_1^T + \dots + \lambda_n v_n v_n^T$*

Rayleigh Quotient *$R(x)=x^T A x / x^T x$, measuring a scaling effect*

Gram Matrix *One of the most common symmetric matrices, covered fully in this chapter's Section 4*

Further Reading

Further Reading · 11.3 Asymmetric Matrix EVD

An asymmetric matrix's eigendecomposition is no longer "rotate $\rightarrow$ scale $\rightarrow$ rotate"

Further Reading · 11.4 The Gram Matrix

This chapter's most common symmetric matrix — an important application of spectral decomposition

Review · 12.2 The Ellipse View of PCA

Spectral decomposition's idea of straightening an ellipse is PCA's geometric foundation

Open-Source Repository

Code and figures for the whole book: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1a Find the spectral decomposition of the real symmetric matrix $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, and explain the result geometrically.

Q2 Modify LA_11_01_01.ipynb to visualize the spectral decompositions of both matrices in Q1 with Python.

Q4 In your own words, explain why spectral decomposition's geometric meaning is "rotate $\rightarrow$ scale $\rightarrow$ rotate".

Q6 Explain why VTv_1 , VTv_2 come out to exactly the standard basis e_1 , e_2 — is this a coincidence?

Q8 Derive why switching a rotated ellipse's coordinate system to $[v_1, v_2]$ straightens it.

Q1b Find the spectral decomposition of the real symmetric matrix $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, and explain the result geometrically.

Q3 Prove: a real symmetric matrix's eigenvalues are always real (never imaginary or complex).

Q5 For this section's matrix A , verify that VT 's determinant is 1, and explain what that means.

Q7 Write the matrix multiplication form of the standard-form ellipse equation $x_1^2/a^2 + x_2^2/b^2 = 1$.

Q9 Explain the geometric meaning of the Rayleigh quotient $R(x) = x^T A x / x^T x$, and its relation to eigenvalues.

Exercises Q10 – Q18

Q10 Prove: eigenvectors of a real symmetric matrix belonging to distinct eigenvalues are always orthogonal.

Q12 Build a 3×3 real symmetric matrix, find its spectral decomposition with `numpy.linalg.eigh`, and check that V is orthogonal.

Q14 Use `matplotlib` to visualize a 2×2 real symmetric matrix's transform of the unit circle, marking the ellipse's semi-axes.

Q16 Prove: a real symmetric matrix A 's trace (sum of main-diagonal entries) equals the sum of all its eigenvalues.

Q18 Explore: if a real symmetric matrix A has a repeated eigenvalue, does spectral decomposition still hold? Are the eigenvectors still unique?

Q11 Write a Python function `spectral_decomp(A)` that returns V , Λ , and verify $A = V\Lambda V^T$.

Q13 Explain why `numpy.linalg.eigh`, rather than `numpy.linalg.eig`, is the better choice for symmetric matrices.

Q15 Prove: a real symmetric matrix A 's determinant equals the product of all its eigenvalues.

Q17 Using the outer-product view, write out a 3×3 real symmetric matrix's decomposition into 3 rank-one matrices.

Exercises Q19 – Q25

Q19 Build a singular (non-invertible) real symmetric matrix, and verify that at least one eigenvalue is 0.

Q21 Write the matrix-product expression for extracting Λ 's row 3, column 2 entry.

Q23 Explain why the unit circle, transformed by a real symmetric matrix A , always yields an ellipse rather than some other curve.

Q25 Preview: the next section covers eigendecomposition of asymmetric matrices — will its eigenvectors still be orthogonal?

Q20 Use Python to verify $A^{-1} = V\Lambda^{-1}V^T$, and compare it against `numpy.linalg.inv(A)`.

Q22 Explore: is a Gram matrix GTG always real symmetric? What special property do its eigenvalues have?

Q24 Explore: in this section's rotated-ellipse equation $2.125x_1^2 + 3.75x_1x_2 + 2.125x_2^2 = 1$, where does the cross term $3.75x_1x_2$ come from?

10 Prompts to Explore with AI

1 Explain what a real symmetric matrix is, and why its eigendecomposition is specifically called spectral decomposition.

2 In geometric language, explain why spectral decomposition can be described as "rotate $\rightarrow$ scale $\rightarrow$ rotate".

3 Explain how spectral decomposition finds a coordinate system that straightens a rotated ellipse into standard form.

4 Derive the full process of transforming the unit circle by matrix A into an ellipse and finding its equation.

5 Explain the definition of the Rayleigh quotient $R(x)=x^T A x / x^T x$, and its relation to a matrix's eigenvalues.

10 Prompts to Explore with AI

6 Explain matrix multiplication's outer-product view, and how it writes $A=V\Lambda V^T$ as a sum of n rank-one matrices.

7 Explain why a real symmetric matrix is invertible only when all its eigenvalues are nonzero, with a proof sketch.

8 Use Python to demonstrate how to extract specific entries from a diagonal matrix Λ and an identity matrix I .

9 Explain the relationship between real symmetric matrices and Gram matrices, and why Gram matrices are often symmetric.

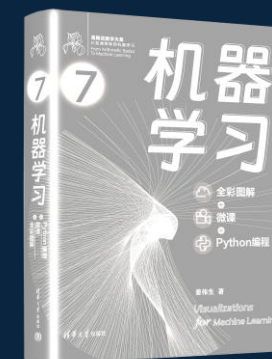
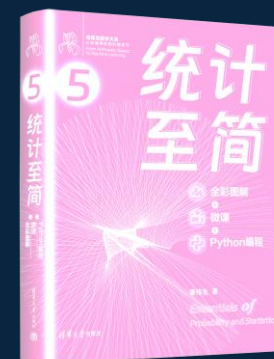
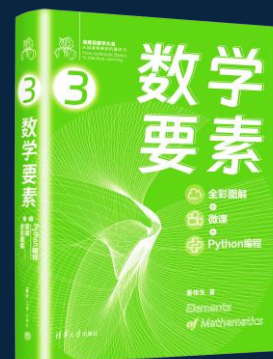
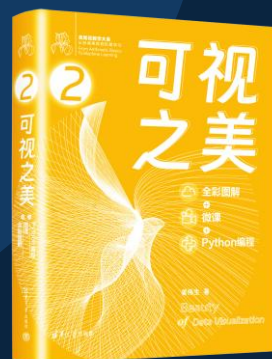
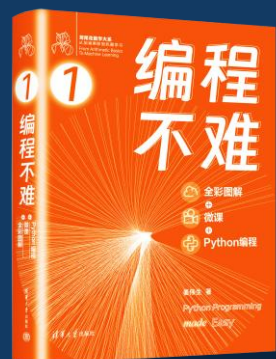
10 Use Python to visualize the three steps of a real symmetric matrix's spectral decomposition: "rotate $\rightarrow$ scale $\rightarrow$ rotate".

数学

数学基础

线性代数

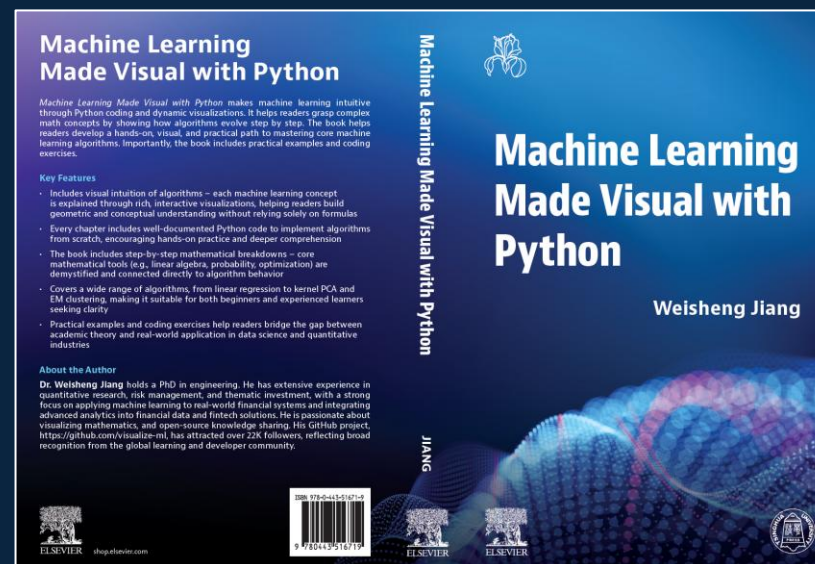
概率统计



Python编程

数据可视化

工具



数据分析
图和网络

回归、降维
分类、聚类

实践